

NVIDIA GSI OPEN HACKATHON 2026

CropGuard AI

Multimodal Agricultural AI Assistant — www.cropguard.in

From crop image to trusted, evidence-based agricultural action

Multimodal diagnosis

LoRA domain adaptation

Enterprise RAG

Team - Multimodel Mavericks



OUR VISION

From Detection to Empowerment - Healthy Crops, Native-Language AI, Stronger Farming Communities, Stronger Nation..



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Special mentoring...

Saurav Mishra, NVIDIA

Agenda

Nine sections — from the farmer's problem to a deployed, evidence-grounded assistant

01 The problem Statement

02 Our Approach

03 NVIDIA MultiModal Architecture & Techstack

04 Fine-Tuning Pipeline

05 Evaluation of Finetuning

06 Evidence and trust

07 Demo Walkthrough

08 Deployment and next steps

09 Experience & Highlights

9 sections • 7-minute walkthrough • recorded live demo included

SECTION 01

The Problem Statement

The real cost isn't the disease — it's the delay.



Diagnosis is uncertain

Similar-looking diseases, image variability, and limited expert access mean symptoms could point to fungal or bacterial disease, nutrient deficiency, insect damage, or environmental stress.



Expertise is always there but not near

With no expert on hand, farmers lean on WhatsApp groups, local input dealers, neighbors, and YouTube — access to a real expert, when needed, is unreliable.



Fragmented knowledge

Useful guidance is distributed across agricultural extension material, research literature, and community sources — not consolidated anywhere a farmer can reliably reach.



A wrong action costs real money

A disease name alone doesn't answer what to do next. Misdiagnosis wastes time and inputs while disease spreads — costing yield, money, and trust.

“My crop doesn’t look healthy. What should I do?”

Plants showing symptoms
5,000 • 500 (10%)

Farmer needs **Fast diagnosis + action** Expert access **Delayed**

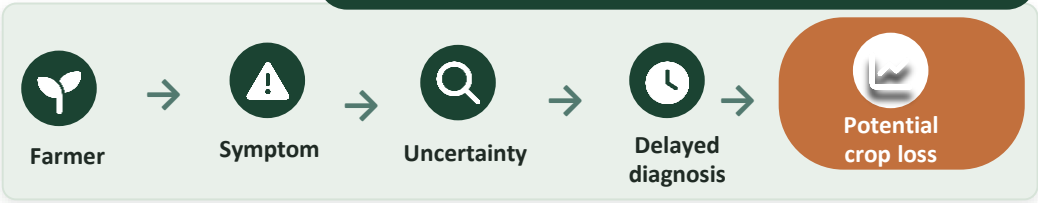
CROP VALUE AT RISK

₹40,000

SECTION 02 — SCALING THE IMPACT

CropGuard at Scale — Assumption: ₹40,000 loss exposure per farm

Farms	Loss exposure	Potential avoided loss at 25%	Potential avoided loss at 50%
1,000	₹4 crore	₹1 crore	₹2 crore
10,000	₹40 crore	₹10 crore	₹20 crore



₹1 crore potential avoided loss — per 1,000 farms, assuming 25% of exposure is avoided

CropGuard objective: Connect visual diagnosis with controlled, retrievable, and evidence-based agricultural guidance — so the farmer’s next action, not just a disease name, is what reaches them.

Our Approach

A three-layer pipeline: vision, reasoning, and evidence.



Layer 1 Vision

Farmer provides a crop image, classified by an NVIDIA vision model.

Tomato Leaf Image

Vision Model

Disease / Symptom Candidates

EXAMPLE OUTPUT

Early Blight	0.91
Septoria Leaf Spot	0.06
Leaf Mold	0.03



Layer 2 Reasoning

Combines the vision result with crop, location, and farmer context.

Crop

Location

Symptoms

Farmer question

Visual diagnosis

SAMPLE CONTEXT

Crop: Tomato
Stage: 45 days
Symptoms: Brown circular lesions
Environment: High humidity
Vision result: Early Blight 91%

MODEL OUTPUT

"The visual symptoms are consistent with early blight."



Layer 3 Evidence

Retrieves supporting agricultural information for the diagnosis.

Diagnosis

Evidence Retrieval

Agricultural Sources

Advisory

WHY IT MATTERS

The advisory reaching the farmer is grounded in verified sources — not a bare disease label.

Each layer maps to an NVIDIA model — see the stack on the next slide.



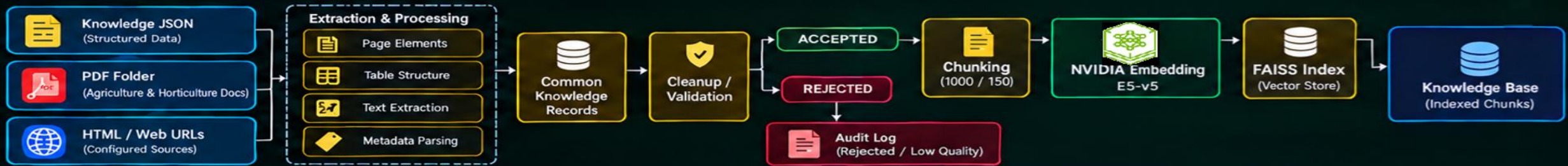
cropguard.in

NVIDIA Multimodal Architecture

ONLINE Farmer Query Pipeline (Real-time Diagnosis & Recommendation)



OFFLINE Knowledge Indexing Pipeline (Build / Update Knowledge Base)



RAG Retrieval Flow (Online)



LLM / Vision Models



Data Processing



Data Store



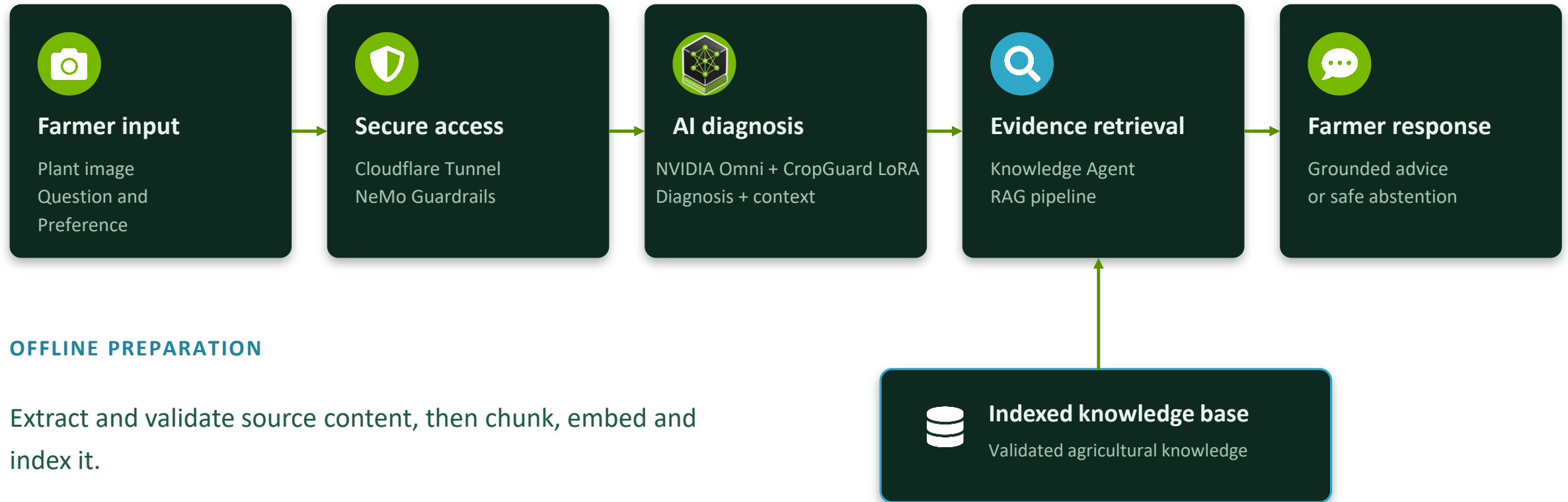
User / Application



Cloudflare Tunnel

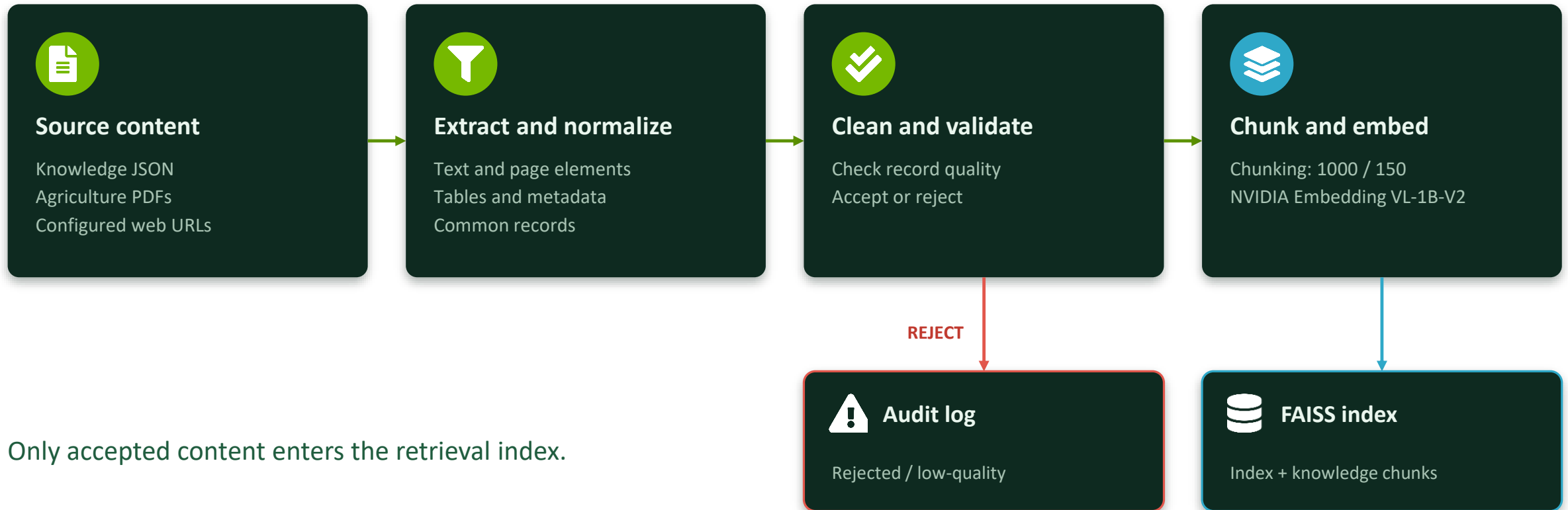
CropGuard architecture

Online diagnosis uses farmer context and retrieved agricultural evidence



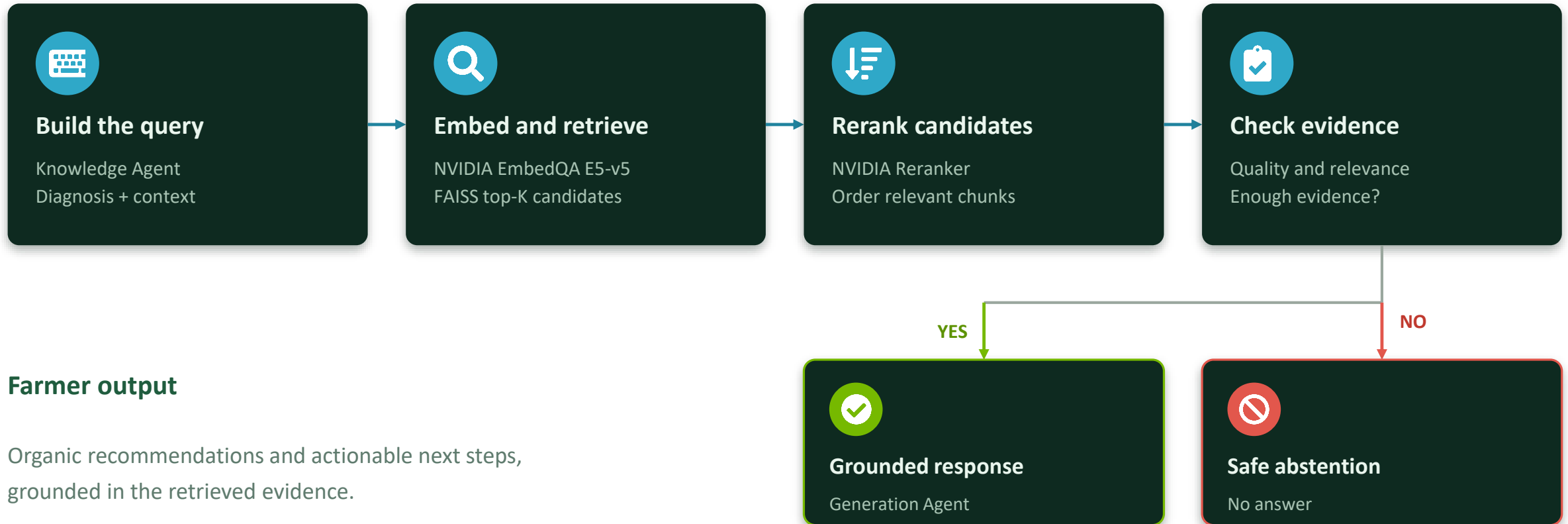
Knowledge indexing pipeline

Offline preparation builds and updates the knowledge base



RAG retrieval and response

Online retrieval checks evidence before generating recommendations



Farmer output

Organic recommendations and actionable next steps,
grounded in the retrieved evidence.

NVIDIA Acceleration, Technology Stack & Final MVP Models

GPU acceleration → NVIDIA technology stack → Diagnostic → Retrieval → Re-ranking → Generation

GPU ACCELERATION — VALIDATED

B300
SXM6 GPU

~275 GB
GPU Memory

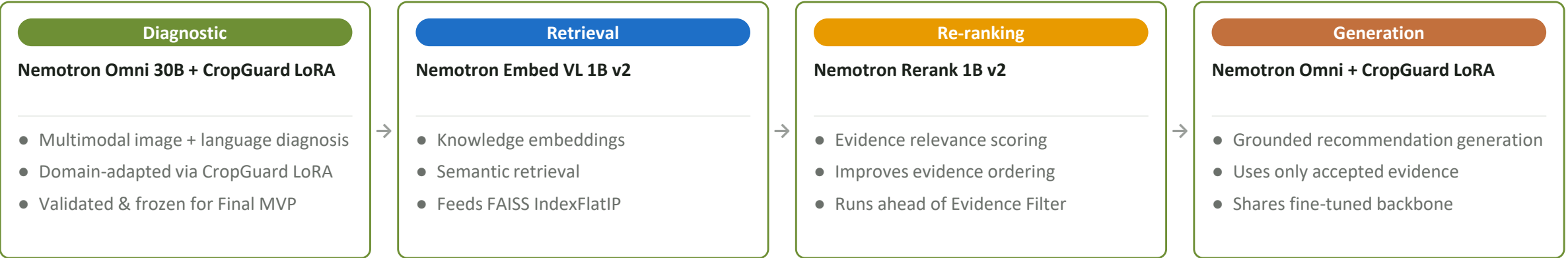
CUDA 13.0
Verified

- **30B-Class VLM Inference**
CPU impractical → GPU (B300, BF16) enables real-time reasoning.
- **LoRA Fine-Tuning**
CPU infeasible at scale → GPU + BF16 + NeMo/PEFT feasible.
- **Embedding Generation**
CPU bottleneck at scale → GPU embedding accelerates RAG.
- **RAG Retrieval & Batch**
CPU limited throughput → GPU + FAISS lowers latency.

END-TO-END TECHNOLOGY STACK

Layer	Technology Stack	Layer	Technology Stack
● Farmer UI	Streamlit	● Embedding	NVIDIA Nemotron Embed VL 1B v2
● Public Access	Cloudflare Tunnel → cropguard.in	● Vector DB / Retrieval	FAISS IndexFlatIP
● Backend API	Python + FastAPI	● Reranking	NVIDIA Nemotron Rerank 1B v2
● AI Orchestration	Python + CropGuard Agents	● Grounding	Evidence Filter + Safe Abstention
● Vision / Reasoning	NVIDIA Nemotron-3-Nano-Omni-30B-A3B-BF16	● Runtime	CUDA + Apptainer + Slurm
● Fine-Tuning	NVIDIA NeMo AutoModel + PyTorch + LoRA/PEFT	● Evaluation	Python + scikit-learn
● Distributed Training	FSDP2 + NVIDIA B300	● E2E Deployment	FastAPI + Streamlit + Cloudflare Tunnel
● Knowledge Base	JSON/JSONL + PDF + Web/HTML sources		

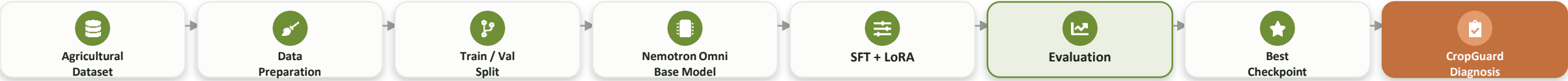
MODELS USED — FINAL MVP PIPELINE



Fine-Tuning Pipeline

Candidate model evaluation → Nemotron Omni selection → Multimodal SFT + LoRA → CropGuard diagnosis

DATA & FINE-TUNING FLOW



DATASETS

PlantVillage

Disease image dataset · 24 disease classes

PlantDoc

Real-world plant images · 17 disease classes

Rice

Rice disease images · 6 disease classes

105,834 SFT training samples | 11,759 validation samples

	MODELS EVALUATED	No. of gpus used	Result
1	Llama 3.2 11B-Vision Baseline / LR behaviour	1 gpu	Discontinued
2	Nemotron Nano V2 Baseline + LoRA evaluation	1 gpu	Evaluated
3	Nemotron Omni 30B Multimodal SFT + LoRA	2 gpu	Selected
4	Cosmos 3 Nano Vision baseline + LoRA evaluation	1 gpu	Evaluated

SFT + LoRA CONFIGURATION

SUPERVISED FINE-TUNING

Recipe: FinetuneRecipeForVLM

Framework: NeMo AutoModel

Task: Multimodal Image-Text SFT

Precision: BF16

Max seq len: 4096

Collator: cropguard_nemotron_collate_fn

LoRA / PEFT

Rank · Alpha: 8 · 32

Learning rate: 1e-4

Optimizer: AdamW

Target: LM linear layers

Frozen: Vision + Audio + Embeddings

LM Head: Excluded

TRAINING SETUP

Epochs: 2

Global batch: 8

Local batch: 4 / GPU

GPUs: 2

Strategy: FSDP2

DP / EP / TP: 2 / 1 / 1

Steps: 26,459

Training loss: 4.055 → 0.0053 (convergence)

Best validation checkpoint: step 22,999

20+ engineering issues resolved — checkpoint conversion, LoRA module remapping, multimodal token handling, custom collator, dataset schema, FSDP2/NCCL, and 2-GPU allocation

- NVIDIA
- Hugging Face
- OpenACC

Evaluation of Finetuning

Cosmos 3 Nano vs. Nemotron Omni 30B — same 100-image benchmark, same LoRA recipe (rank 8, alpha 32)



EVALUATION SET

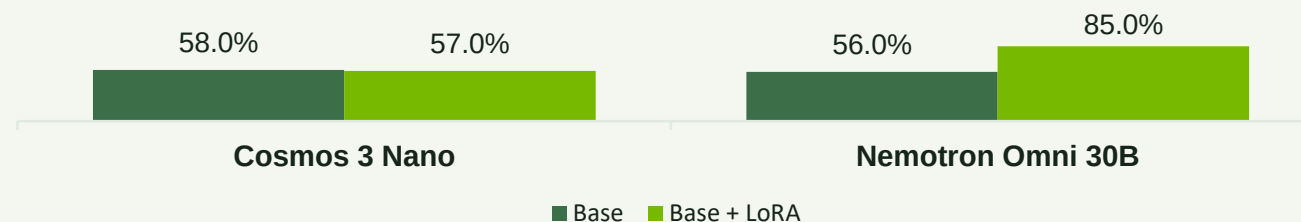
Balanced dataset of **100 images**



WHY OMNI WAS SELECTED

- +29.0 pt accuracy gain from LoRA, vs. a -1.0 pt net change for Cosmos 3
- All 5 disease classes improved; Cosmos + LoRA collapsed 2 classes to 0% recall
- Balanced precision/recall across the board post-LoRA (86% / 85% macro avg)
- Faster after LoRA: 12.9s / image vs. 24.3s baseline, and fewer truncated responses (2 vs. 11)

ACCURACY — BASE vs. BASE + LoRA (100-image test set)



Cosmos 3: **-1.0 pt** Nemotron Omni: **+29.0 pt**

NEMOTRON OMNI — LoRA EVALUATION KPIs

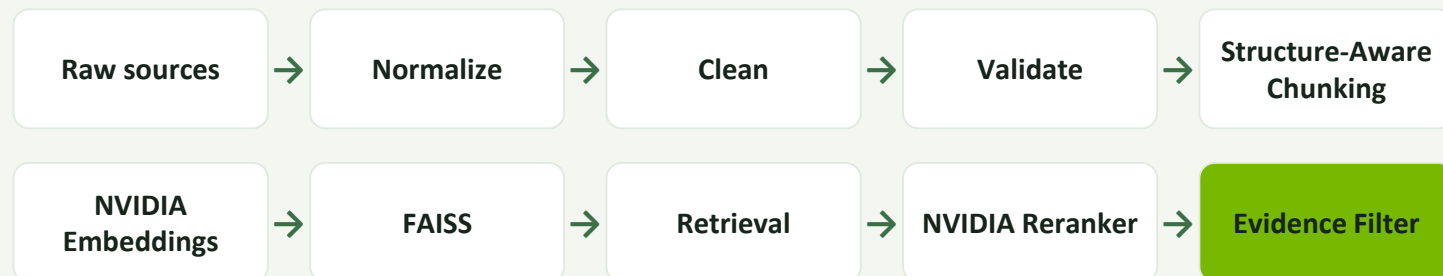
Metric	Base	Base + LoRA	Δ
Accuracy	56.0%	85.0%	+29.0
Precision (macro)	49.6%	85.7%	+36.0
Recall (macro)	56.0%	85.0%	+29.0
F1 (macro)	51.6%	84.7%	+33.1

32 images improved · 3 regressed · 53 both correct · 12 both incorrect

Evidence and trust

Evidence is a control layer that constrains retrieval and generation.

KNOWLEDGE PIPELINE



configured web sources: TNAU, Cornell, Json, PDF

Rejected knowledge is never indexed, retrieved or reranked.



If evidence is insufficient, **CropGuard abstains instead of fabricating a treatment recommendation.**

Demo Walkthrough

Image-first agricultural assistant — shown as a 60–90 sec flow: upload → diagnosis → confidence / symptoms → recommendation → evidence → traceability.

DEMO NARRATIVE

- 1 Upload crop image
- 2 Ask farmer question
- 3 Select language + preference
- 4 Show crop / disease + confidence + visual symptoms
- 5 Show grounded recommendation
- 6 Show evidence and source traceability
- 7 Show healthy / negative / insufficient-evidence behavior



VALIDATED E2E CASE

Corn + Northern Leaf Blight

High-confidence diagnosis → sufficient evidence → grounded recommendation

TRUST Source / chunk traceability shown on screen.

SAFETY Insufficient eligible evidence → no unsupported treatment.

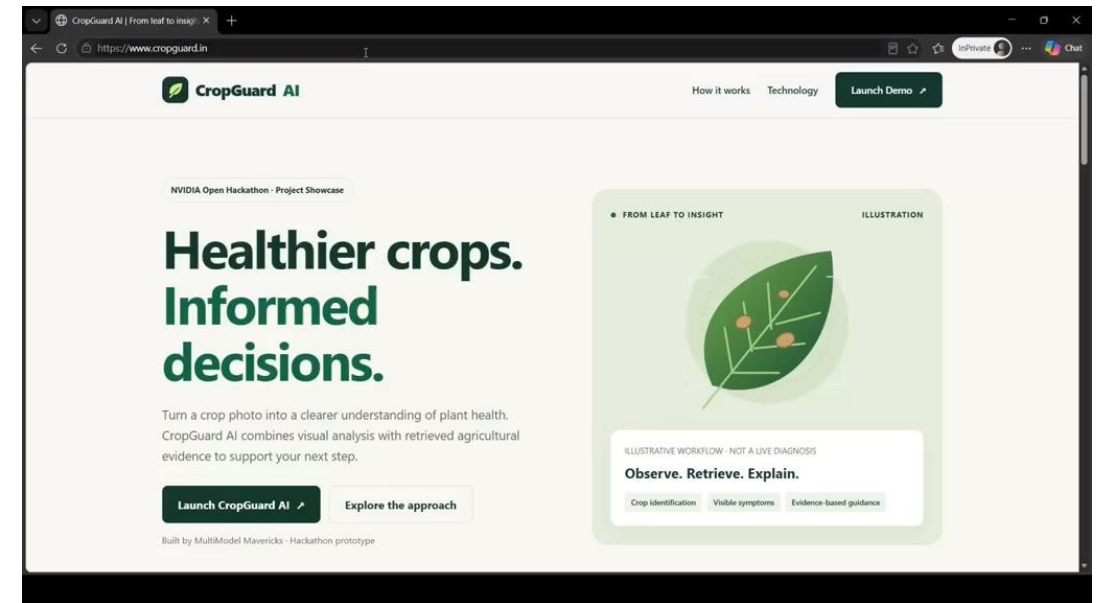
USE CASE SCENARIOS

Healthy crop identification

Tomato :Disease identification and Recommendation

Corn: Disease identification and Recommendation

Guardrails - Evidence



Deployment and next steps

From NVIDIA GPU infrastructure to a live farmer-facing web experience

CURRENT DEPLOYMENT



NVIDIA GPU infrastructure



Apptainer



Nemotron Omni + LoRA



FastAPI



Streamlit



Cloudflare named tunnel



Farmer web experience

www.cropguard.in

demo.cropguard.in

DEPLOYMENT CONSIDERATIONS

NVIDIA NIM

Containerization

Model versioning

Enhanced Monitoring

Improved Latency

Scalability

Data privacy



FUTURE CAPABILITIES



Regional voice interface



Multilingual Input



Weather-aware diagnosis



Crop-stage awareness



Disease progression tracking



Field-level monitoring



Satellite / drone imagery



Expert escalation



Offline / edge inference

Experience & Highlights

From three MVPs to one deployed platform — what we built, and why it worked.

WHAT WE EXPERIENCED

- ✓ **Built and deployed 3 MVPs** using different architectures and technology stacks, enabling rapid experimentation and comparison.
- ✓ **Trained and evaluated 4 models** and used the evaluation results to select the model for the final CropGuard MVP.
- ✓ **Worked through the complete AI lifecycle** data prep → multimodal SFT → LoRA → evaluation → RAG → E2E deployment.
- ✓ **Deployed the farmer-facing app at cropguard.in** exposed through a Cloudflare Tunnel for secure external access and live demonstration.
- ✓ **Connected every piece end-to-end** AI backend, farmer-facing UI, RAG pipeline and external web access, into one demonstrable MVP.

WHAT MADE IT BETTER

- ★ **Valuable guidance from the NVIDIA mentor** helped us define a clear objective for each MVP instead of solving everything in one iteration.
- ★ **Multiple MVPs, evaluated in a short window** each iteration produced concrete, comparable technical learning.
- ★ **An iterative approach** surfaced the strengths, limitations and integration challenges of different models and architectures.
- ★ **Hands-on depth** with NVIDIA Nemotron Omni, multimodal processing, LoRA, FSDP2, GPU-based training/inference and evidence-grounded RAG.
- ★ **A real deployment, not just a demo** moving from experimentation to actual E2E deployment made the learning significantly more valuable.

Was It Worth It?

Looking back at the build — and ahead at where CropGuard goes next.



Was It Worth It?

Absolutely. The power of NVIDIA's compute resources and development environment fueled our curiosity, accelerated experimentation, and inspired us to think beyond the initial prototype toward bigger, more ambitious goals.



Will You Continue Development?

Definitely. This experience demonstrated what is possible when the right AI ecosystem, compute power, and experimentation environment come together.

NEXT STEPS & FUTURE PLANS

1

Larger, more
diverse datasets



2

Better disease
detection accuracy



3

Expanded RAG
knowledge base



4

Native-language,
context-aware advice



5

Scalable, sustainable
crop-health platform

RESOURCES & SUPPORT

Mentorship and continued access to powerful NVIDIA compute resources will be critical to accelerate model training, experimentation, validation, and scaling.

IDEAL COMPUTE SIZE

8-GPU • 1–2 TB RAM

A high-memory, 8-GPU NVIDIA system would give the next phase a strong environment for larger-model fine-tuning, multimodal experimentation, RAG development, and rapid iteration.



Thank You

Acknowledgement & Next Opportunities

This work was completed in part at the GSI Open Hackathon, part of the Open Hackathons program. **The team acknowledges OpenACC and NVIDIA** for the opportunity, infrastructure, and technical ecosystem that enabled this work.

Future opportunities: continue CropGuard evaluation and development, and share results through technical presentations, demonstrations, papers or internal innovation forums — with the required Open Hackathons / OpenACC acknowledgement.



Appendix

The information beyond we present

Key Engineering Challenges Resolved

Nemotron Omni deployment • Multimodal SFT • LoRA • Distributed training • Evaluation

Severity	Area	Engineering Issue	Solution / Resolution
CRITICAL	Nemotron Omni Deployment	Custom Nemotron Omni architecture required the correct loading path	Loaded local Omni snapshot with <code>trust_remote_code=True</code> and validated the custom architecture
CRITICAL	Multimodal Processing	Image tokens were not created correctly with direct text-only processor invocation	Used explicit image + text conversation format and <code>apply_chat_template()</code>
CRITICAL	VLM Data Pipeline	SFT records were not directly compatible with the Nemotron Omni VLM schema	Built CropGuard conversation-based image + text SFT records
CRITICAL	Custom VLM Collator	Image paths required preprocessing before NVIDIA's Omni collator	Implemented <code>cropguard_nemotron_collate_fn</code> : RGB conversion + 512×512 normalization
CRITICAL	LoRA Integration	LoRA target module paths did not match the actual Nemotron hierarchy	Remapped <code>language_model.model.layers.*</code> → <code>language_model.backbone.layers.*</code>
CRITICAL	LoRA Checkpoint	Existing adapter could not directly attach after module-path resolution	Created remapped adapter while preserving the original checkpoint
HIGH	LoRA Validation	Needed to prove the remapping was lossless	0 missing • 0 extra • 0 shape errors • 0 value mismatches • max diff = 0
HIGH	Distributed Training	30B multimodal model required multi-GPU training	Configured FSDP2 + 2 GPUs • DP=2 • EP=2 • TP=1 • CP=1
HIGH	Multimodal Inference	Multimodal metadata caused failures when incorrectly passed through generation	Investigated and isolated the custom Nemotron generation interface
HIGH	Controlled Evaluation	Base vs LoRA visual improvement required identical evaluation conditions	Same images + prompts + preprocessing + labels for controlled comparison

NVIDIA-focused highlights: Nemotron Omni multimodal integration • Custom VLM collator • LoRA architecture remapping • Lossless adapter validation • FSDP2 2-GPU SFT • Controlled Base vs LoRA evaluation

Key Engineering Challenges Resolved Contd.,

Inference stability • Evaluation • Knowledge validation • RAG • E2E application

Severity	Area	Engineering Issue	Solution / Resolution
HIGH	Inference Stability	Individual evaluation images could stall generation and block the complete evaluation run	Added per-image generation timeout and controlled evaluation handling
HIGH	Evaluation Scope	Leaf Blast was not recognized reliably in the current MVP evaluation	Excluded Leaf Blast from the final benchmark and retained a balanced 5-class evaluation scope
HIGH	VLM Output Parsing	Free-form model responses were not directly suitable for automated classification metrics	Added diagnosis extraction and canonical disease-label normalization
HIGH	Knowledge Validation	Invalid or rejected knowledge records could contaminate the retrieval pipeline	Added validation before chunking and indexing; rejected records are not indexed
HIGH	RAG Retrieval	Vector retrieval alone could return semantically similar but lower-quality evidence	Added FAISS candidate retrieval followed by Nemotron reranking and evidence filtering
HIGH	Grounded Recommendation	Recommendations required sufficient supporting evidence before generation	Added evidence sufficiency checking with a safe abstention path when evidence is insufficient
HIGH	E2E Service Orchestration	API, UI and external access components required coordinated startup and lifecycle control	Created a unified CropGuard lifecycle script for start, stop, restart, status and clean operations
HIGH	GPU Container Execution	Model inference required consistent GPU visibility between the DGX host and Apptainer	Standardized Apptainer --nv execution and validated CUDA/GPU availability before inference

NVIDIA-focused highlights: Inference stability • Controlled evaluation scope • VLM output normalization • Knowledge validation • Reranked evidence • Grounded generation • E2E service orchestration

Dataset composition

PlantVillage + PlantDoc + Rice1426 → unified CropGuard schema



PLANTVILLAGE

9 crops · 24 classes

Apple: apple scab, black rot, cedar apple rust

Cherry: powdery mildew

Corn: cercospora leaf spot, common rust, northern leaf blight

Grape: black rot, esca, leaf blight

Orange: citrus greening

Peach: bacterial spot

Pepper: bacterial spot

Potato: early blight, late blight

Strawberry: leaf scorch

Tomato: bacterial spot, early blight, late blight, leaf mold, mosaic virus, septoria leaf spot, spider mites, target spot, yellow leaf curl virus



PLANTDOC

8 crops · 19 classes

Apple: apple scab, black rot, cedar apple rust

Cherry: powdery mildew

Corn: cercospora leaf spot, common rust, northern leaf blight

Grape: black rot

Peach: no disease label

Pepper: bacterial spot

Potato: early blight, late blight

Squash: powdery mildew

Tomato: bacterial spot, early blight, late blight, leaf mold, mosaic virus, septoria leaf spot, spider mites, yellow leaf curl virus

Excludes 4 malformed records (healthy, and mold/rust/scab → UNKNOWN) — kept out of the treatment KB.



RICE1426

1 crop · 6 classes

Rice: bacterial leaf blight, brown spot, leaf blast, scald, sheath blight, healthy

Cleanest of the three sources — every record maps to a well-formed croAp / label pair.

Key Engineering Fixes

Getting LoRA fine-tuning and multimodal training working on Nemotron Omni



Base vs. LoRA module-path mismatch

A module-path mismatch, not a weight or shape mismatch.

```
LoRA expected language_model.model.layers.*  
Base exposed language_model.backbone.layers.*
```

Fix: remapped adapter targets to the Base hierarchy, then validated every tensor.

Missing / extra tensors	Shape / value mismatches	Max tensor diff
0 / 0	0 / 0	0



Custom VLM collator

Bridges image and conversational records into Nemotron Omni's expected tensors.

```
AssertionError: No image tokens found  
in input_ids
```

Fix: build the multimodal conversation first, then run the processor and chat template so image tokens stay aligned.

Bridges PlantVillage, PlantDoc and Rice into the Nemotron Omni training pipeline.